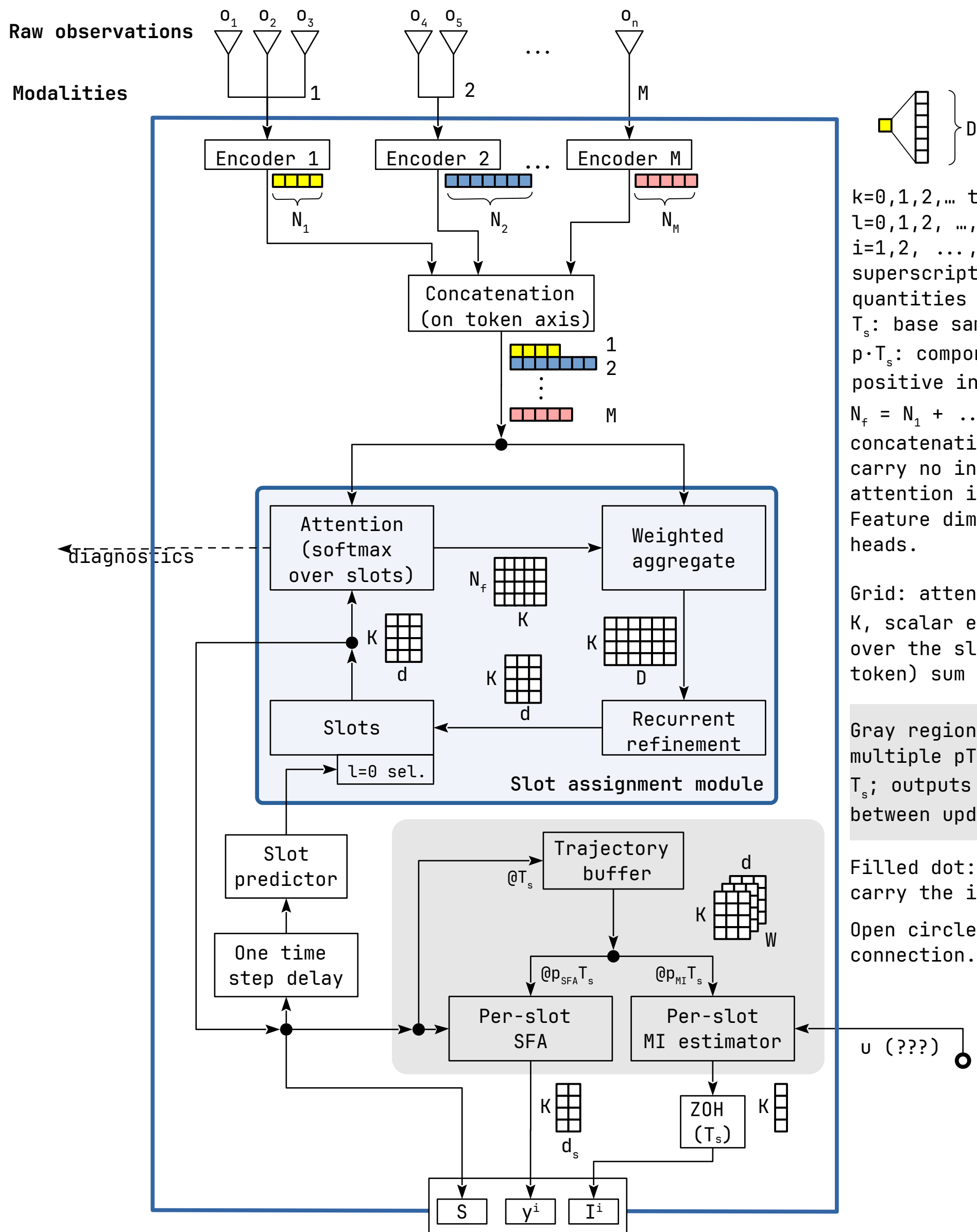


Layer 1 internal structure (working sketch)



$k=0,1,2,\dots$ time step (frame)
 $l=0,1,2,\dots, L$ iterations per frame
 $i=1,2,\dots, K$ slot index;
 superscript i marks per-slot
 quantities (y^i, I^i)
 T_s : base sample period (one frame tick)
 $p \cdot T_s$: component sample period, p a
 positive integer (p_{SFA}, p_{MI})
 $N_f = N_1 + \dots + N_M$ after
 concatenation. Token and row order
 carry no information: downstream
 attention is permutation invariant.
 Feature dimension D is common to all
 heads.

Grid: attention weight matrix A , $N_f \times K$, scalar entries; softmax normalized over the slot axis; rows (one per token) sum to 1.

Gray region: sample period an integer multiple pT_s of the base frame period T_s ; outputs are zero-order held between updates. White region: $p=1$.

Filled dot: fan-out; all branches carry the identical signal.

Open circle: unresolved boundary connection.

Implementation notes (working)

1. Delay state is the slot matrix S ($K \times d$). At $K = 8$, $d = 128$, fp32, this is approximately 4KB: fits in on-chip SRAM, no DRAM round trip required.
 2. Double buffering required: $S[k]$ remains immutable while $S[k+1]$ is computed; bank swap at the tick boundary, no in-place update.
 3. Single-writer, multi-reader contract: the encoder writes once per tick; SFA, the MI estimator, the L2 handoff, and T read the same frozen $S[k]$.
 4. The delay defines the sample period: all in-loop computation must complete within one tick; the encoder dominates the budget, the delay memory is negligible.
 5. Fixed-rate execution is mandatory: SFA slowness statistics and T's fixed-step dynamics are corrupted by tick jitter; fixed-rate scheduling or per-sample timestamps required.
 6. The delay block is not the storage element: replay/minibatch buffers (MINE/InfoNCE) and SFA covariance statistics belong to the estimators and are megabyte-scale; drawn separately to prevent misreading.
 7. Device boundary risk: if the encoder runs on an accelerator and the estimators on host, the device-host transfer sits at the delay boundary; transfer latency eating the tick budget, not bandwidth, is the risk.
 8. Wire rates: @-annotated; unannotated wires transfer at the base period T_s .
- ZOH (Zero-Order Hold): holds its input between updates and republishes it at T_s (rate converter from $p \cdot T_s$ to T_s).
9. Blue region: slot assignment module; iterated $l = 0 \dots L$ within each frame tick.
 10. Attention softmax over the slot axis implements competition: rows of A , one per token, sum to 1. The aggregate renormalizes A 's columns over tokens and computes a weighted mean, so each slot's update is a convex combination of its claimed tokens and does not grow with the number claimed.
 11. The token wire carries F ; the learned value projection v is applied inside the block. The aggregate lands in projection width D ; Recurrent refinement is the D -to- d bridge into slot space.
 12. Holds the current estimate of the K slot vectors; refined L times per frame; initialized each frame from the previous frame's prediction.
 13. $l = 0$ sel.: input source selector for the Slots register. At $l = 0$ of each frame it admits the slot predictor's output (the previous frame's prediction); for $l = 1 \dots L$ it admits the Recurrent refinement output.